

APSMO 2012 Set 4

1 Problems

Problem 1.1 Consecutive numbers are counting numbers that follow in order. For example: 3, 4, 5, 6 are four consecutive numbers.

Which four consecutive numbers have a sum equal to 70?

Problem 1.2 $X * Y$ represents the fraction $\frac{(X+Y)}{(X-Y)}$.

Express the number represented by $12 * (7 * 5)$ in its simplest form.

Problem 1.3 There are five Mondays in a certain month.

If three of the five Mondays occur on even days, which day of the week is the 11th day of the month?

Problem 1.4 In the division problem below, different letters represent different digits. AB is an even 2-digit number. EEE is a 3-digit number.

Find the 2-digit number AB .

$$\begin{array}{r} AB \\ M \overline{) EEE} \end{array}$$

Problem 1.5 A rectangular prism that is 4cm by 6cm by 8cm is painted on all six faces.

The prism is cut into cubes that measure 2cm on each side.

How many of these cubes have only one face painted?